

SOME BASIC CONCEPTS OF CHEMISTRY

Class 11 · Maharashtra Board · MHT-CET Foundation · Chapter 1

MISMATHE Study Sheet · Mindmap + Formulas + Must-Remember · Test: 5 July

MINDMAP — The 6 Branches

1 · Nature of Matter

- Matter = has mass + occupies space
- States: Solid → Liquid → Gas (interconvertible by T, P)
- Pure substance:** Element (1 atom type) / Compound (fixed ratio)
- Mixture:** Homogeneous (uniform) / Heterogeneous (non-uniform)

2 · Measurement

- 7 SI base units: m, kg, s, A, K, mol, cd
- Density = mass / volume**
- K = °C + 273.15**
- Sig figs: non-zero & in-between zeros count; leading zeros don't
- ×, ÷ → fewest sig figs | +, - → fewest decimals

3 · Laws of Chemical Combination (5) + Dalton ★ exam favourite

- Conservation of Mass** (Lavoisier) — mass is neither created nor destroyed in a reaction.
 - Definite Proportions** (Proust) — a compound always has same elements in same mass ratio (H₂O always 1:8 H:O).
 - Multiple Proportions** (Dalton) — masses of one element combining with a fixed mass of another are in a simple whole-number ratio (CO vs CO₂ → O = 1:2).
 - Gay-Lussac's Law** — gases react in simple whole-number volume ratios.
 - Avogadro's Law** — equal volumes of all gases (same T, P) contain equal number of molecules.
- Dalton's Atomic Theory:** matter = indivisible atoms; same-element atoms identical; combine in whole-number ratios; atoms neither created nor destroyed.

4 · Atomic & Molecular Masses

- 1 u = 1/12 mass of a C-12 atom
- Avg atomic mass** = $\Sigma(\text{isotope mass} \times \text{abundance})$
Cl = $(35 \times 0.75) + (37 \times 0.25) = 35.5$ u
- Molecular mass = Σ atomic masses (H₂O = 18 u)
- Formula mass = ionic compounds (NaCl = 58.5 u)

5 · Mole Concept ★★★ THE HEART

- 1 mole = 6.022×10^{23} particles (N^T)
- = molar mass in grams = 22.4 L gas at STP
- Particles = $n \times 6.022 \times 10^{23}$

★ MASTER FORMULA ★ $\text{moles (n)} = (\text{given mass} / \text{molar mass}) = (\text{no. of particles} / 6.022 \times 10^{23}) = (\text{volume at STP} / 22.4 \text{ L})$

6 · Formulae, Stoichiometry & Concentration

- % element** = $(\text{mass of element} / \text{molar mass}) \times 100$
- Empirical formula = simplest whole-number atom ratio
- Molecular formula** = $(\text{Empirical})_n$, $n = \text{mol. mass} / \text{emp. mass}$
- Limiting reagent** = reactant that finishes first → decides product amount (convert each to moles, compare with equation ratio)
- Molarity (M)** = $\text{mol solute} / \text{L solution}$
- Molality (m)** = $\text{mol solute} / \text{kg solvent}$
- Mole fraction** = $n^o / n^{o\text{H}_{\text{tot}}}$
- Mass %** = $(\text{mass solute} / \text{mass solution}) \times 100$
- ppm** = $(\text{mass solute} / \text{mass solution}) \times 10^6$

MUST-REMEMBER FORMULA SHEET

write these on one page

1	$n = \text{mass} / \text{molar mass} = N / (6.022 \times 10^{23}) = V(\text{STP}) / 22.4$
2	Density = mass / volume
3	$K = ^\circ\text{C} + 273.15$ $^\circ\text{F} = (9/5)^\circ\text{C} + 32$
4	Average atomic mass = $\Sigma(\text{isotopic mass} \times \text{fractional abundance})$
5	% element = $(\text{mass of element} / \text{molar mass of compound}) \times 100$
6	Molecular formula = $(\text{Empirical formula})_n$; $n = \text{molecular mass} / \text{empirical mass}$
7	Molarity (M) = moles of solute / volume of solution (L)
8	Molality (m) = moles of solute / mass of solvent (kg)
9	Mole fraction = moles of component / total moles

🕒 WHAT THE EXAM LOVES — Marks-Grabbers

- **Mole conversions** (mass ↔ moles ↔ particles ↔ volume) — guaranteed numerical, master this first.
- **Molarity / Molality** sums — easy marks once the formula is memorized.
- **Limiting reagent** — one medium-hard numerical, very commonly asked.
- **Law of Multiple Proportions** — short theory + ratio question.
- **Significant figures** — 1-mark MCQ, basically free marks.
- **Empirical** → **Molecular formula** — classic 3-mark question.

🕒 TONIGHT'S 3-HOUR PLAN

B1	Foundations: matter, units, sig figs, 5 laws + Dalton (45 min)
B2	★ MOLE CONCEPT + mass/mole/particle/volume numericals (45 min)
B3	Empirical/molecular formula, stoichiometry, limiting reagent (45 min)
B4	Concentration terms + 10-question closed-book self-test (20 min)

"Connect first, then improve." — You're not an average student. You're a Class 11 kid who studies on his off day. | MISMATHE · Your MHT-CET Mentor